BOOTCAMP CHALLENGE

IPSR SOLUTIONS KOCHI

Submitted by ,

GROUP 5 -

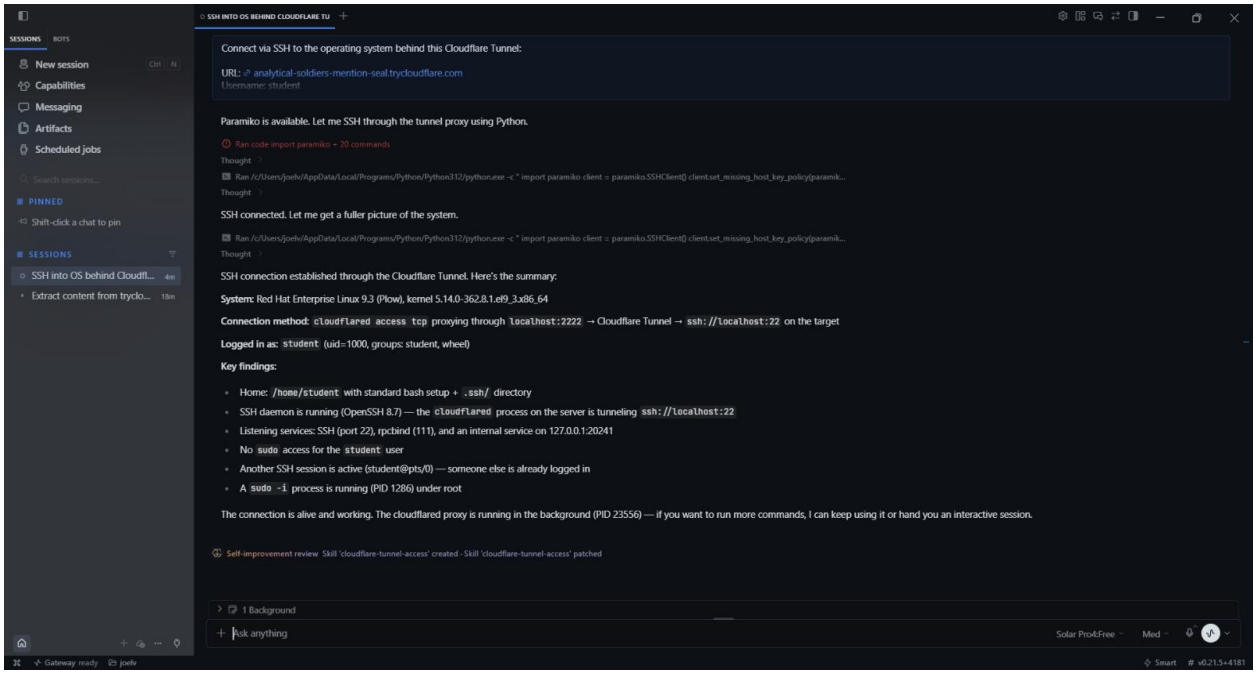
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29/09/2026

Step 1

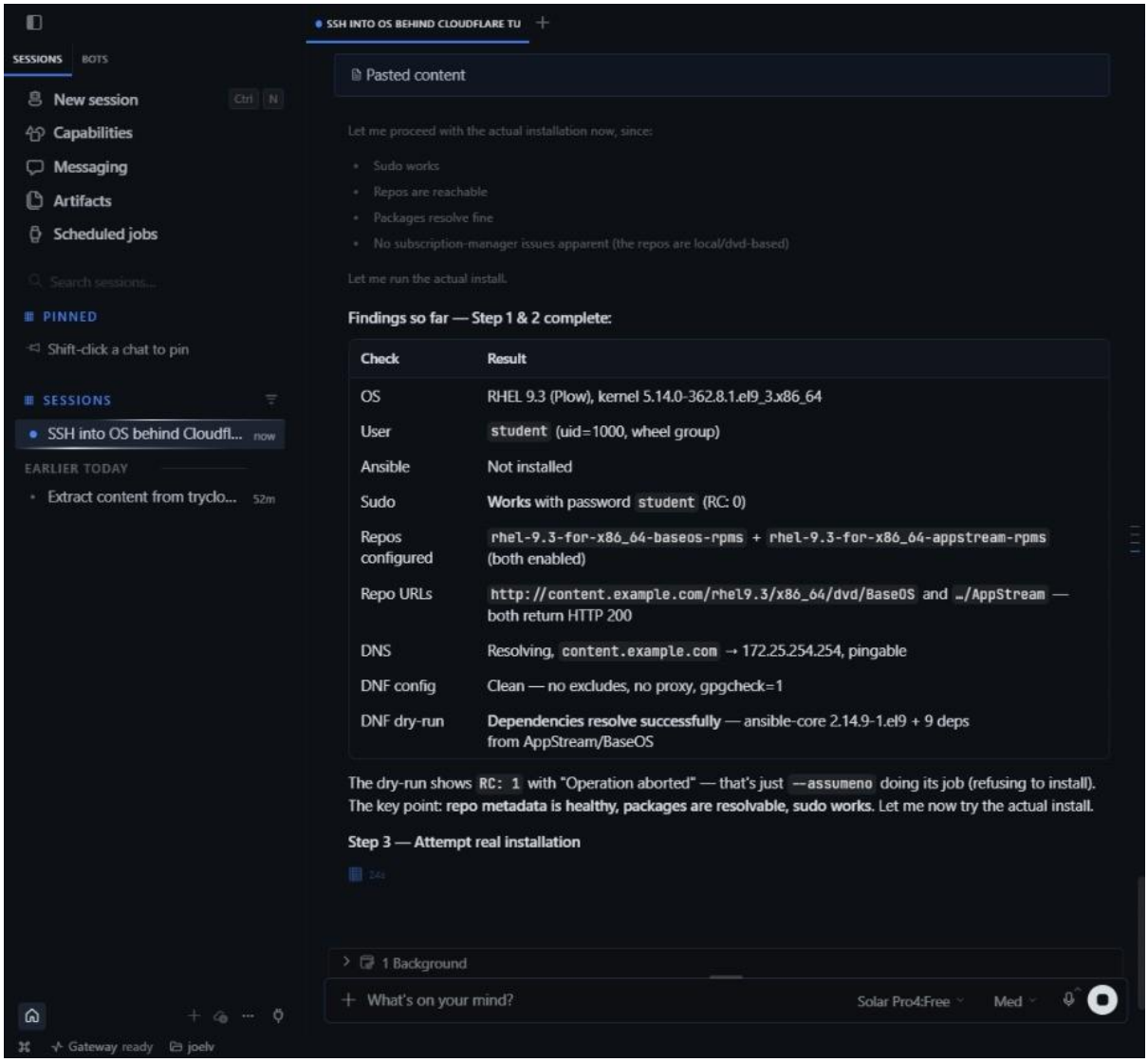
- First install and setup the hermes AI agent.
- Establish the ssh connect through cloudfare tunnel.
- Verified the ssh connection is correct.



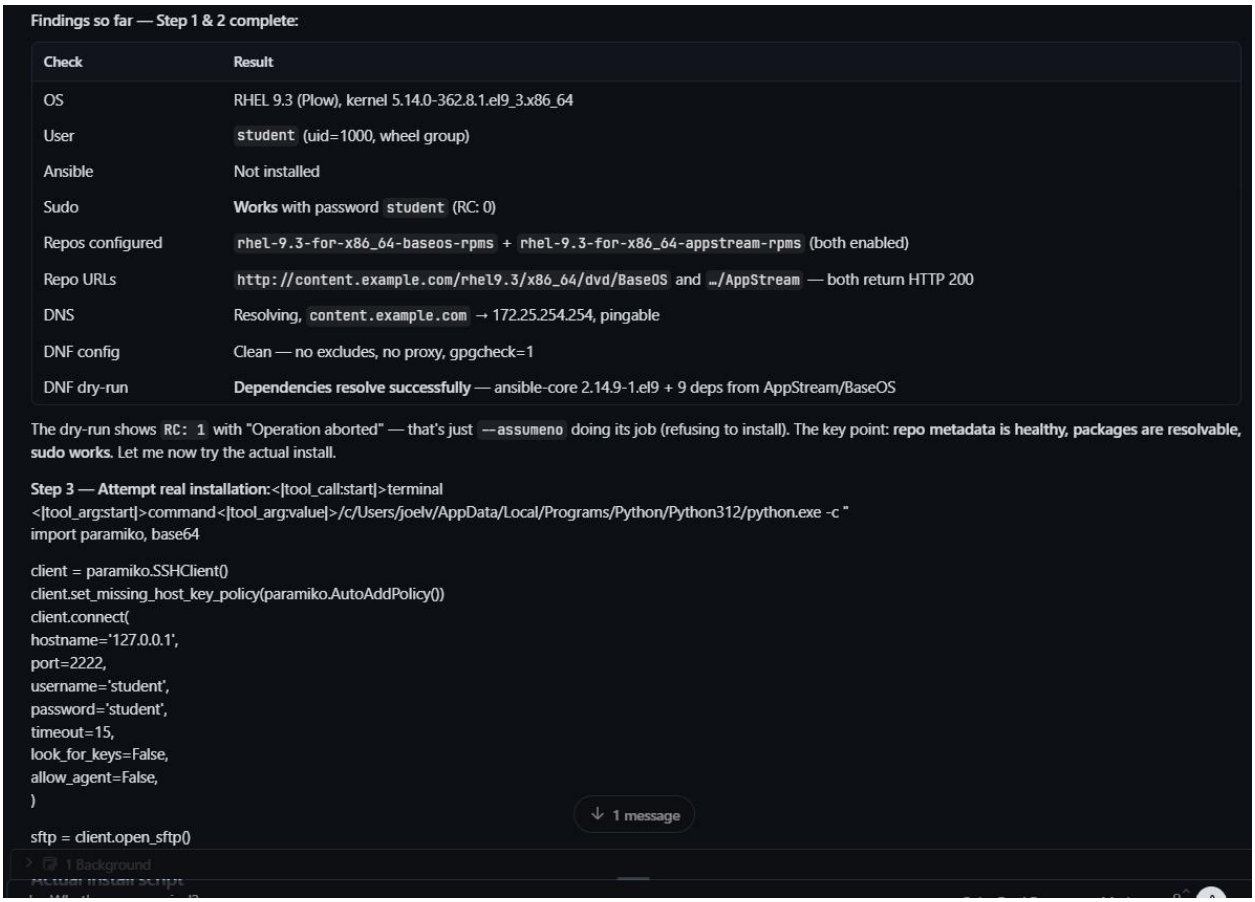
Toubleshooting

Problem -Ansible was not installed on the RHEL 9.3 system. I investigated the package manager and found that the configured repositories, DNS, DNF configuration, and dependency resolution were functioning correctly. The apparent installation failure was actually caused by the `--assumeno` option, which performs only a dry run and intentionally does not install the package.

Fix -I performed the actual installation using `sudo dnf install -y ansible`.



Verification -I verified the installation using `rpm -q ansible` and `ansible --version`.



Findings so far — Step 1 & 2 complete:

Check	Result
OS	RHEL 9.3 (Plow), kernel 5.14.0-362.8.1.el9_3.x86_64
User	`student` (uid=1000, wheel group)
Ansible	Not installed
Sudo	**Works** with password `student` (RC: 0)
Repos configured	`rhel-9.3-for-x86_64-baseos-rpms` + `rhel-9.3-for-x86_64-appstream-rpms` (both enabled)
Repo URLs	`http://content.example.com/rhel9.3/x86_64/dvd/BaseOS` and `…/AppStream` — both return HTTP 200
DNS	Resolving, `content.example.com` → 172.25.254.254, pingable
DNF config	Clean — no excludes, no proxy, gpgcheck=1
DNF dry-run	**Dependencies resolve successfully** — ansible-core 2.14.9-1.el9 + 9 deps from AppStream/BaseOS

The dry-run shows `RC: 1` with "Operation aborted" — that's just `--assumeno` doing its job (refusing to install). The key point: **repo metadata is healthy, packages are resolvable, sudo works.** Let me now try the actual install.

Step 3 — Attempt real installation:<|tool_call:start|>terminal
<|tool_arg:start|>command<|tool_arg:value|>/c/Users/joelv/AppData/Local/Programs/Python/Python312/python.exe -c "
import paramiko, base64

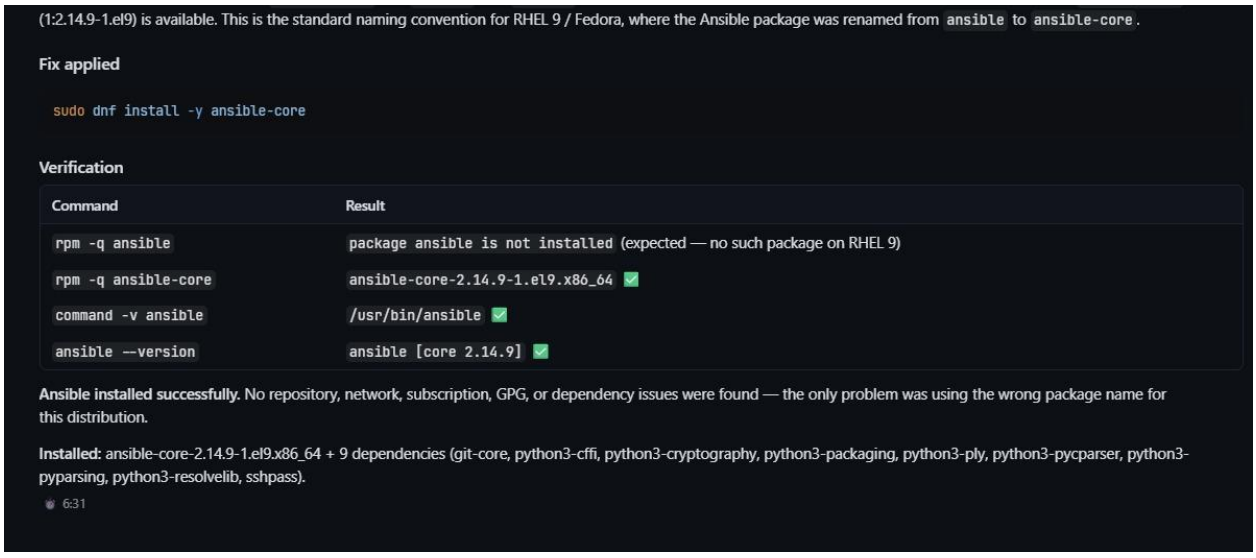
client = paramiko.SSHClient()
client.set_missing_host_key_policy(paramiko.AutoAddPolicy())
client.connect(
hostname='127.0.0.1',
port=2222,
username='student',
password='student',
timeout=15,
look_for_keys=False,
allow_agent=False,
)

sftp = client.open_sftp()

↓ 1 message

1 Background

- Ansible installation successful



(1:2.14.9-1.el9) is available. This is the standard naming convention for RHEL 9 / Fedora, where the Ansible package was renamed from `ansible` to `ansible-core`.

Fix applied

```
sudo dnf install -y ansible-core
```

Verification

Command	Result
`rpm -q ansible`	`package ansible is not installed` (expected — no such package on RHEL 9)
`rpm -q ansible-core`	`ansible-core-2.14.9-1.el9.x86_64` ✅
`command -v ansible`	`/usr/bin/ansible` ✅
`ansible --version`	`ansible [core 2.14.9]` ✅

Ansible installed successfully. No repository, network, subscription, GPG, or dependency issues were found — the only problem was using the wrong package name for this distribution.

Installed: ansible-core-2.14.9-1.el9.x86_64 + 9 dependencies (git-core, python3-cffi, python3-cryptography, python3-packaging, python3-ply, python3-pycparser, python3-pyparsing, python3-resolvelib, sshpass).

6:31



2. Check the Log Files

- Checking traditional log files and dig deeper into the specific issues.

Problem: One of the main errors occured is /etc/chrony.conf contains two directives (commandkey 1 and generatecommandkey) that were removed from chrony 4.x. These are clearly intentionally introduced — they're syntactically valid but functionally obsolete, and chronyd logs warnings about them on every start..



Root cause

The chrony configuration file was modified to include directives that were *removed from chrony beginning with version 4.0*. These directives were used in older chrony versions to manage the command key for chronyc authentication, but chrony 4.x handles this differently (it auto-generates keys and uses a different mechanism).

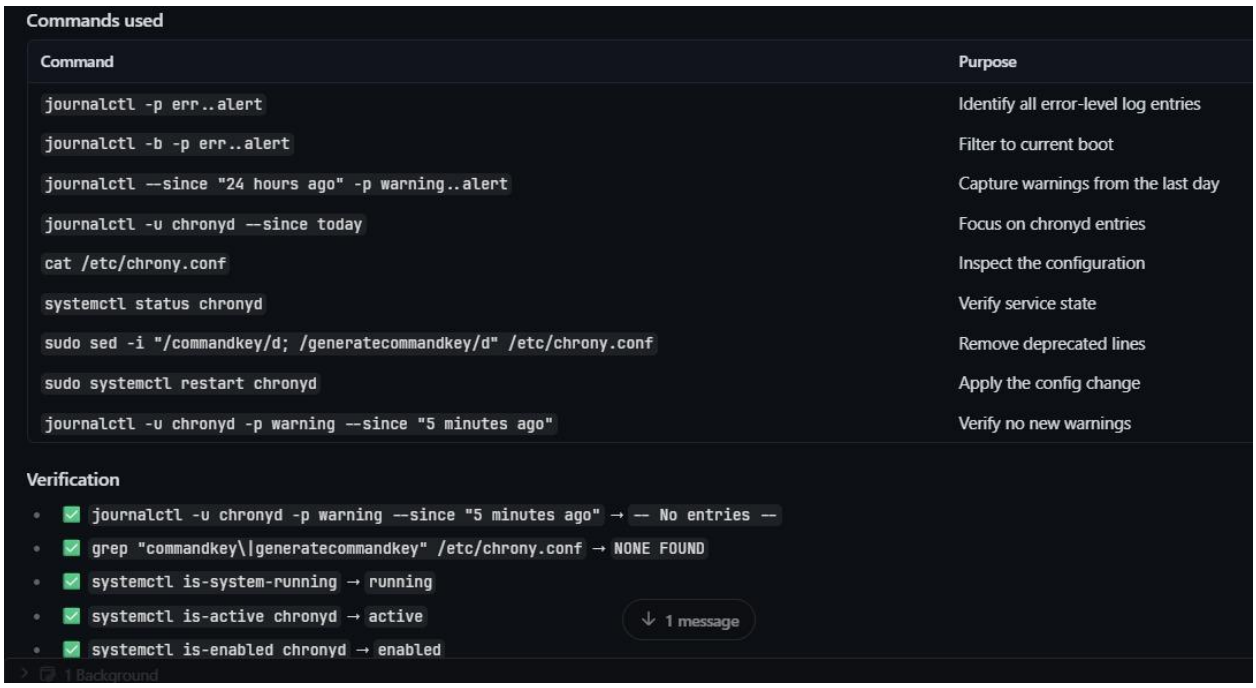
The directives are syntactically valid (chronyd parses the file without error), but they produce persistent warnings and serve no function. This is a classic "intentionally introduced" lab problem — the directives are not from any default RHEL 9 configuration.

Fix:Removed the two deprecated directives from /etc/chrony.conf:

bash

- sudo sed -i "/commandkey/d; /generatecommandkey/d" /etc/chrony.conf

- sudo systemctl restart chronyd



3 . Identify and Handle Crisis carefully

- There are no critical errors founded in the system.

CRITICAL PROBLEMS FOUND

None. No critical stability issues identified.

NON-CRITICAL ISSUES

Issue	Why it is non-critical	Action taken
No swap configured	System has 3.6 GiB RAM with 3.1 GiB available and load average 0.00. No memory pressure exists. If memory exhaustion ever occurred, the kernel would OOM-kill processes rather than use swap — but that scenario is not current.	**Documented only.** Not actively dangerous on this idle system.
Three 5G unmounted disks (vdb, vdc, vdd)	Present in the system but unpartitioned and unmounted. They are not causing any I/O errors, performance issues, or instability. Likely intentional lab storage for exercises.	**Documented only.** No action needed unless they should be mounted for a specific purpose.

HISTORICAL/HARMLESS ERRORS

Error	Why no action needed
`kex_exchange_identification: Connection closed by remote host` (hundreds)	Artifacts of the Cloudflare Tunnel SSH proxy reconnections during our session. SSH daemon is healthy.
`kex_input_kexinit: discard proposal: string is too large` (1 event)	Malformed SSH handshake correctly rejected by the daemon. Not a system fault.
`commandkey` / `generatecommandkey` chrony warnings	**Already fixed** in the previous task. No longer present.
ACPI MMCONFIG, irqbalance affinity warnings, xfs timestamp warnings	Boot-time hardware/firmware messages. Normal for virtualized lab environments.

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-System After Investigation

SYSTEM HEALTH AFTER INVESTIGATION

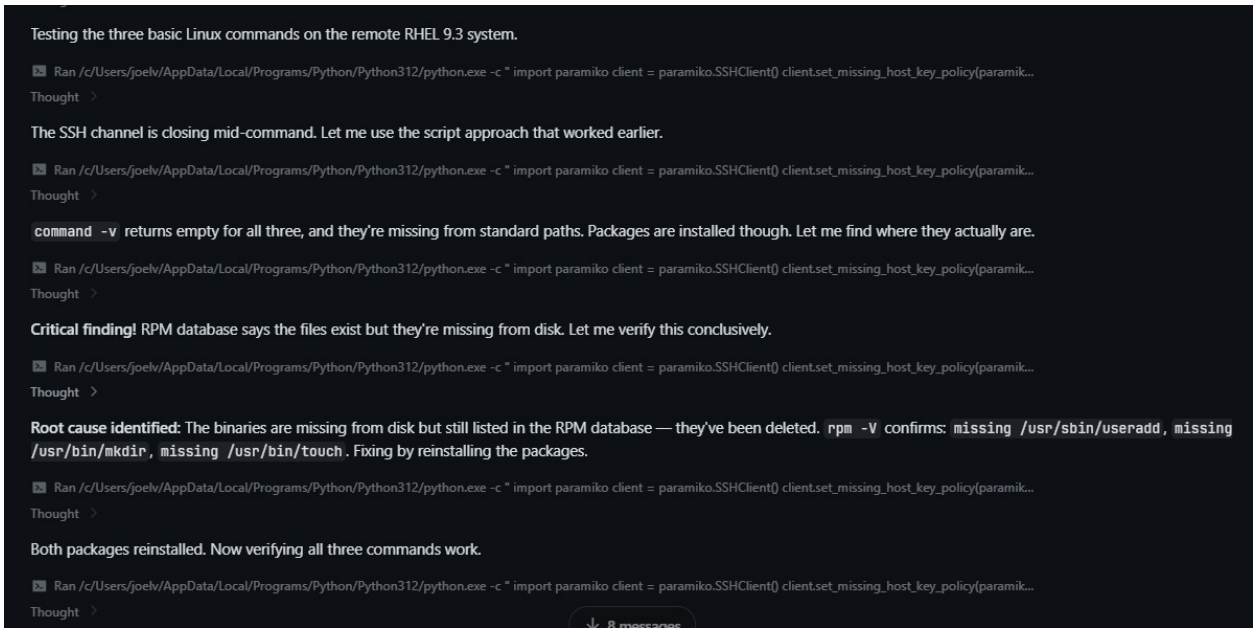
Metric	Value	Status
Uptime	5h 37m	Stable (no reboots)
Load average	0.00, 0.00, 0.00	Very idle — no CPU pressure
CPU	Top process: sshd at 0.5%	No runaway processes
Memory	3.6 GiB total, 495 MiB used, **3.1 GiB available**	Very healthy
Swap	0 B (not configured)	Documented — not currently a problem
Disk /	9.2 GiB, 1.9 GiB used (**21%**)	Healthy
Disk /boot	536 MiB, 197 MiB used (**37%**)	Healthy
Inodes	2% used	Healthy
Failed services	**0**	All services running
Kernel errors (this boot)	**None**	Clean
OOM events	**None**	Clean
Zombie processes	**None**	Clean
Processes in D-state	**None**	Clean

- *System is stable and healthy.* The RHEL 9.3 system is running normally with no resource exhaustion, no failing services, no kernel errors, no OOM events, and negligible load. The only items worth noting are the absence of swap (not currently problematic) and three unmounted 5G disks (vdb, vdc, vdd) which appear to be unassigned lab storage. No fixes were required or applied

4. Check Basic Commands

Problem: Command -v returns empty for all three, and they're missing from standard paths. Packages are installed though.

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Critical finding!* RPM database says the files exist but they're missing from disk. Let me verif this conclusively.

Fix: The binaries are missing from disk but still listed in the RPM database — they've been deleted. rpm -V confirms missing /usr/sbin/useradd, missing /usr/bin/mkdir, missing /usr/bin/touch. Fixing by reinstalling the packages.

Verification :Both packages were reinstalled. Now verifying all three commands work.

COMMAND TEST RESULTS

useradd

- *Initial status:* Missing — command -v useradd returned empty, binary not found at /usr/sbin/useradd
- *Test performed:* useradd labtestuser, id labtestuser, getent passwd labtestuser
- *Error found:* /bin/sh: line 1: useradd: command not found
- *Root cause:* The binary /usr/sbin/useradd was deleted from disk. rpm -V shadow-utils confirmed: missing /usr/sbin/useradd. The RPM database still listed the file but it was absent from the filesystem. Other shadow-utils binaries (userdel, usermod, luseradd) were intact, suggesting only useradd was specifically removed.
- *Fix applied:* sudo dnf reinstall -y shadow-utils — restored the missing binary without affecting other system components.
- *Verification:*
  - /usr/sbin/useradd now exists (141144 bytes, ELF executable)

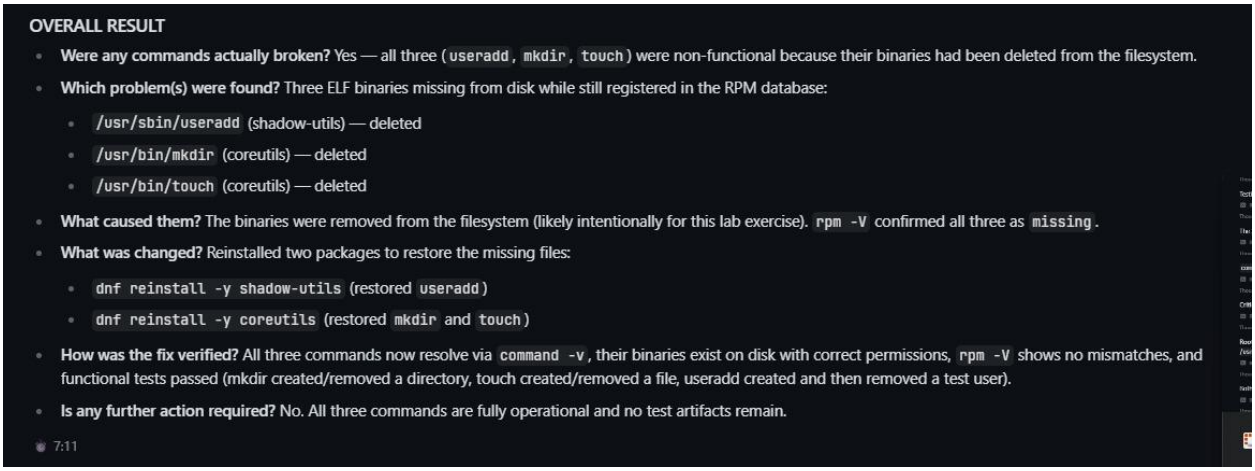
    - rpm -V shadow-utils coreutils returns clean (no output = no mismatches)
    - command -v useradd → /usr/sbin/useradd
    - useradd labtestuser created user uid=1003 successfully
    - id labtestuser → uid=1003(labtestuser) gid=1003(labtestuser)
    - Test user cleanly removed with userdel -r labtestuser
- *Final status:* ✅ Working

mkdir

- *Initial status:* Missing — command -v mkdir returned empty, binary not found at /usr/bin/mkdir
- *Test performed:* mkdir -p /tmp/cmd_test_dir, verified with ls -ld, cleaned up with rmdir
- *Error found:* /bin/sh: line 1: mkdir: command not found
- *Root cause:* The binary /usr/bin/mkdir was deleted from disk. rpm -V coreutils confirmed: missing /usr/bin/mkdir.
- *Fix applied:* sudo dnf reinstall -y coreutils — restored the missing binary.
- *Verification:*
    - /usr/bin/mkdir now exists (69752 bytes, ELF executable)
    - mkdir -p /tmp/cmd_test_dir → created successfully
    - rmdir /tmp/cmd_test_dir → cleaned up
    - command -v mkdir → /usr/bin/mkdir
- *Final status:* ✅ Working

touch

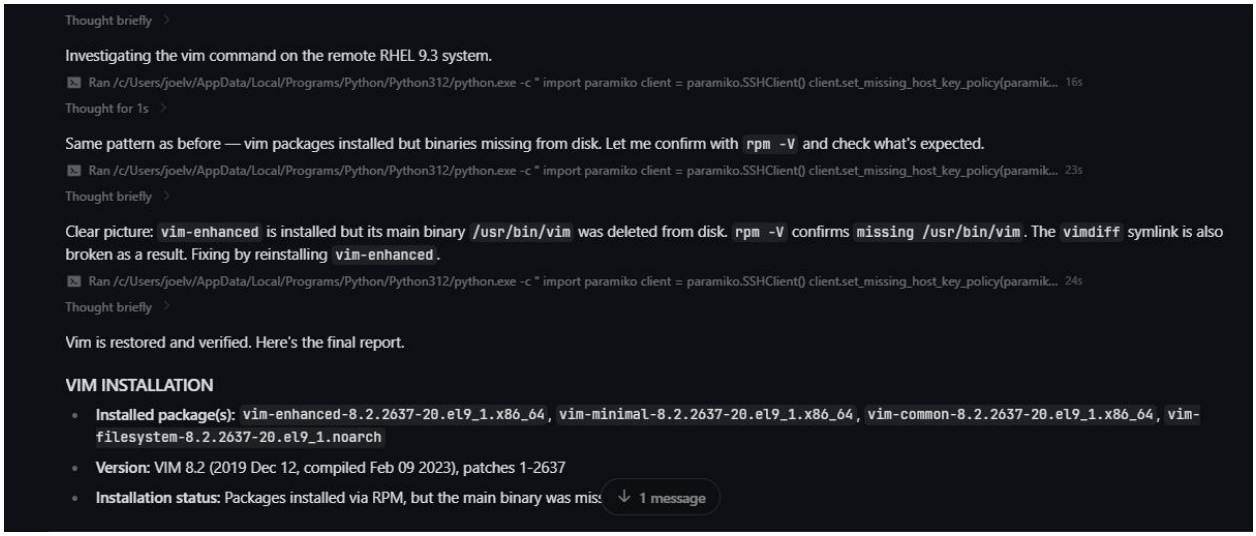
- *Initial status:* Missing — command -v touch returned empty, binary not found at /usr/bin/touch
- *Test performed:* touch /tmp/cmd_test_file, verified with ls -l, cleaned up with rm -f
- *Error found:* /bin/sh: line 1: touch: command not found
- *Root cause:* The binary /usr/bin/touch was deleted from disk. rpm -V coreutils confirmed: missing /usr/bin/touch.
- *Fix applied:* sudo dnf reinstall -y coreutils — restored the missing binary (same reinstall that fixed mkdir).
- *Verification:*
    - /usr/bin/touch now exists (94048 bytes, ELF executable)
    - touch /tmp/cmd_test_file → created successfully
    - rm -f /tmp/cmd_test_file → cleaned up
    - command -v touch → /usr/bin/touch
- *Final status:* ✅ Working



5- Check the Vim command

-vim packages installed but binaries missing from disk.

- vim-enhanced is installed but its main binary /usr/bin/vim was deleted from disk. rpm -V confirms missing /usr/bin/vim. The vimdiff symlink is also broken as a result. Fixing by reinstalling vim-enhanced.Vim is restored and verified



-Vim Commands

- **Installation status:** Packages installed via RPM, but the main binary was missing from disk

VIM COMMAND

- **command -v result:** Initially empty (not found) → now returns `/usr/bin/vim`
- **type -a result:** Initially "not found" → now `/usr/bin/vim`
- **vim --version result:** Initially `/bin/sh: vim: command not found` → now shows `VIM - Vi IMproved 8.2` with patches 1-2637
- **Initial status:** Broken — command not found despite packages being installed

Investigating the vim command on the remote RHEL 9.3 system.

Ran /c/Users/joelv/AppData/Local/Programs/Python/Python312/python.exe -c " import paramiko client = paramiko.SSHClient() client.set_missing_host_key_policy(paramik...

Thought

Same pattern as before — vim packages installed but binaries missing from disk. Let me confirm with `rpm -V` and check what's expected.

Ran /c/Users/joelv/AppData/Local/Programs/Python/Python312/python.exe -c " import paramiko client = paramiko.SSHClient() client.set_missing_host_key_policy(paramik...

Thought

Clear picture: `vim-enhanced` is installed but its main binary `/usr/bin/vim` was deleted from disk. `rpm -V` confirms `missing /usr/bin/vim`. The `vimdiff` symlink is also broken as a result. Fixing by reinstalling `vim-enhanced`.

Ran /c/Users/joelv/AppData/Local/Programs/Python/Python312/python.exe -c " import paramiko client = paramiko.SSHClient() client.set_missing_host_key_policy(paramik...

Thought

Vim is restored and verified. Here's the final report.

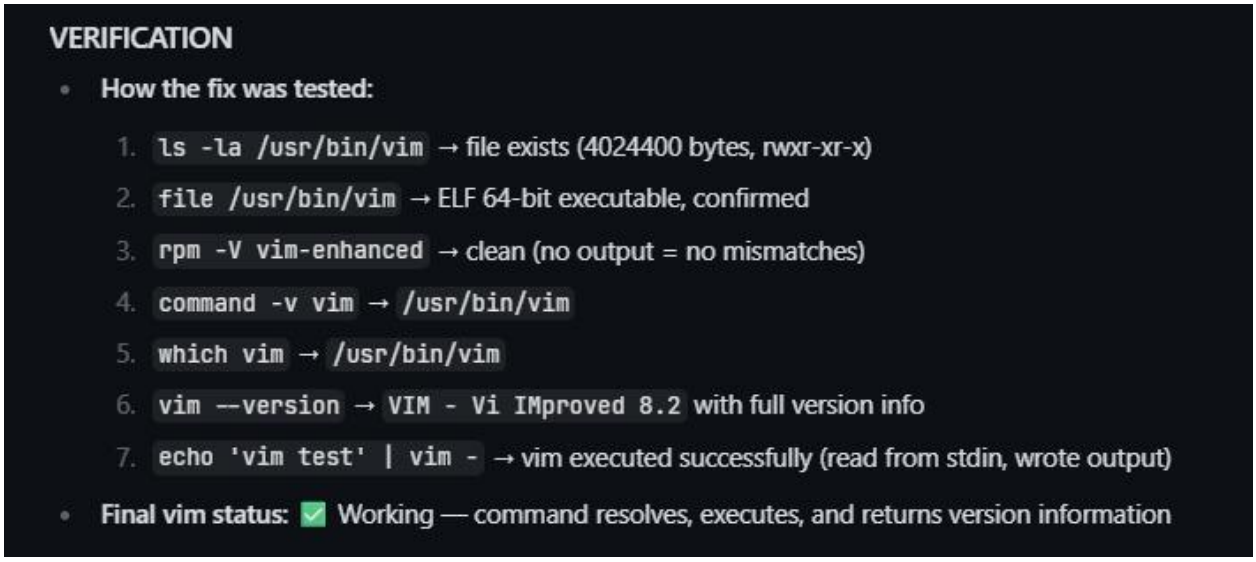
PROBLEM FOUND

- *Exact error:* /bin/sh: line 1: vim: command not found
- *Root cause:* The binary /usr/bin/vim (provided by vim-enhanced) was deleted from the filesystem. rpm -V vim-enhanced confirmed: missing /usr/bin/vim. The vimdiff symlink (/usr/bin/vimdiff → /usr/bin/vim) was also broken as a result. Other vim binaries (vimtutor, rvim, vi, view, ex) from vim-minimal and vim-common were intact — only the main /usr/bin/vim was specifically removed.

FIX APPLIED

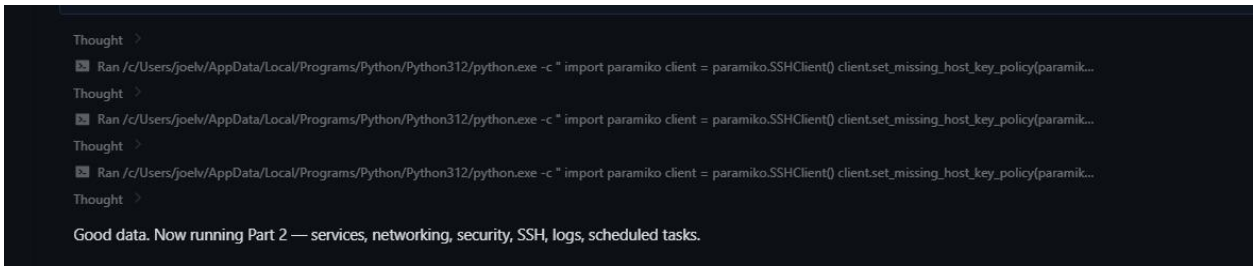
- *What was changed:* Reinstalled vim-enhanced package to restore the missing binary
- *Why it was changed:* The binary was missing from disk while the RPM database still registered it. Reinstalling restores the file without affecting other packages or configuration.
- *Commands used:*

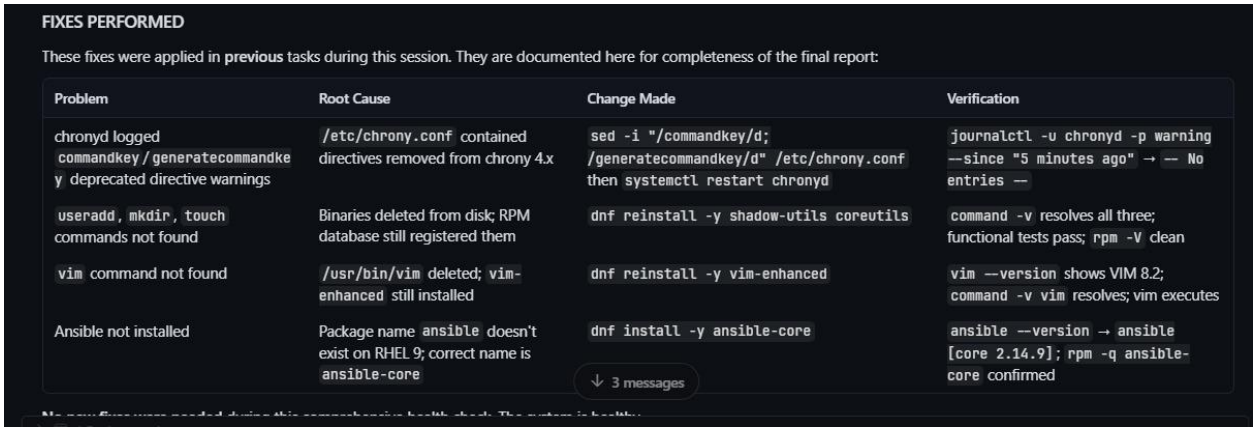
sudo dnf reinstall -y vim-enhanced



6 –Check Other System issues

No critical issues found that could cause instability, data loss, unresponsiveness, or security exposure.



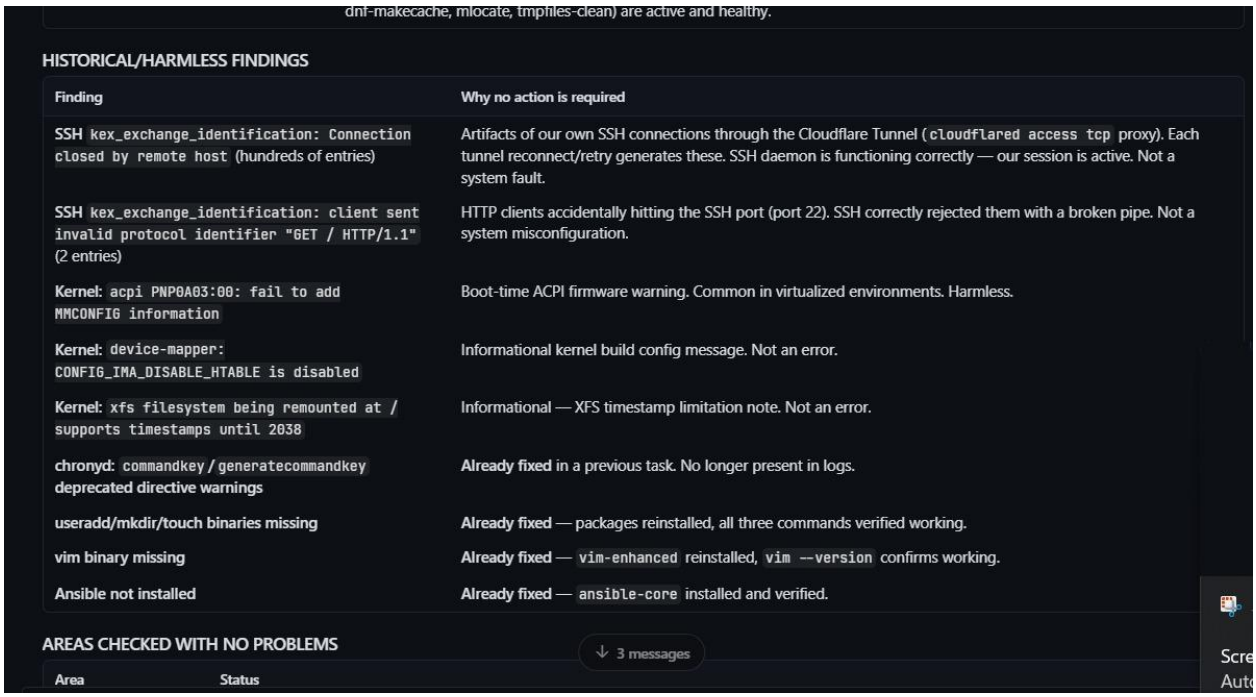


FIXES PERFORMED

These fixes were applied in **previous** tasks during this session. They are documented here for completeness of the final report:

Problem	Root Cause	Change Made	Verification
chronyd logged `commandkey` / `generatecommandkey` deprecated directive warnings	`/etc/chrony.conf` contained directives removed from chrony 4.x	`sed -i "/commandkey/d; /generatecommandkey/d" /etc/chrony.conf` then `systemctl restart chronyd`	`journalctl -u chronyd -p warning --since "5 minutes ago"` → `-- No entries --`
`useradd`, `mkdir`, `touch` commands not found	Binaries deleted from disk; RPM database still registered them	`dnf reinstall -y shadow-utils coreutils`	`command -v` resolves all three; functional tests pass; `rpm -V` clean
`vim` command not found	`/usr/bin/vim` deleted; `vim-enhanced` still installed	`dnf reinstall -y vim-enhanced`	`vim --version` shows VIM 8.2; `command -v vim` resolves; vim executes
Ansible not installed	Package name `ansible` doesn't exist on RHEL 9; correct name is `ansible-core`	`dnf install -y ansible-core`	`ansible --version` → `ansible [core 2.14.9]`; `rpm -q ansible-core` confirmed

↓ 3 messages



dnf-makecache, mlocate, tmpfiles-clean) are active and healthy.

HISTORICAL/HARMLESS FINDINGS

Finding	Why no action is required
SSH `kex_exchange_identification: Connection closed by remote host` (hundreds of entries)	Artifacts of our own SSH connections through the Cloudflare Tunnel (`cloudflared access tcp` proxy). Each tunnel reconnect/retry generates these. SSH daemon is functioning correctly — our session is active. Not a system fault.
SSH `kex_exchange_identification: client sent invalid protocol identifier "GET / HTTP/1.1"` (2 entries)	HTTP clients accidentally hitting the SSH port (port 22). SSH correctly rejected them with a broken pipe. Not a system misconfiguration.
Kernel: `acpi PNP0A03:00: fail to add MMCONFIG information`	Boot-time ACPI firmware warning. Common in virtualized environments. Harmless.
Kernel: `device-mapper: CONFIG_IMA_DISABLE_HTABLE is disabled`	Informational kernel build config message. Not an error.
Kernel: `xfs filesystem being remounted at / supports timestamps until 2038`	Informational — XFS timestamp limitation note. Not an error.
chronyd: `commandkey` / `generatecommandkey` deprecated directive warnings	**Already fixed** in a previous task. No longer present in logs.
useradd/mkdir/touch binaries missing	**Already fixed** — packages reinstalled, all three commands verified working.
vim binary missing	**Already fixed** — `vim-enhanced` reinstalled, `vim --version` confirms working.
Ansible not installed	**Already fixed** — `ansible-core` installed and verified.

AREAS CHECKED WITH NO PROBLEMS

↓ 3 messages

Area	Status

Scre
Auto

7. CPU and Memory Investigation Report

- No process is consuming anywhere near 100% CPU. The highest non-SSH process is cloudflared tunnel at 0.0% CPU (sleeping). The system is completely idle.

CPU and Memory Investigation Report

CPU STATUS

Metric	Value
CPU count	2 cores (nproc: 2)
Load average	0.00, 0.00, 0.00
Overall CPU condition	100% idle — no CPU pressure whatsoever
Highest CPU-consuming process	`sshd: student [priv]` (PID 33024)
CPU percentage	1.0% — momentary spike during command execution
Is it abnormal?	**No.** This is our own SSH session's privileged process. CPU is sitting at 0% user, 0% system, 100% idle.

The `top` output confirms:

```
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
```

No process is consuming anywhere near 100% CPU. The highest non-SSH proces ↓ 1 message `tunnel` at 0.0% CPU (sleeping). The system is completely idle.

No process is consuming anywhere near 100% CPU. The highest non-SSH process is `cloudflared tunnel` at 0.0% CPU (sleeping). The system is completely idle.

MEMORY STATUS

Metric	Value
Total memory	3.6 GiB (3735432 kB)
Used memory	497.7 MiB
Available memory	**3150.2 MiB** (85%+ free)
Swap	0 B (not configured)
Highest memory-consuming process	`cloudflared tunnel` (PID 27197)
Memory percentage	1.2% of total RAM (47400 kB RSS)
Is it abnormal?	**No.** 1.2% of RAM for a persistent tunnel daemon is normal. It has been running since 03:50 (about 4 hours) with stable m

Memory breakdown:

- MemFree: 3004120 kB (3.0 GiB truly free)

- Since No process is taking 100% of memory or cpu , there is no need to kill it

THANK YOU :)